

* Session 3 *

→ Linux history

→ ① Make code small is beautiful

② Do one thing Well

③ Every thing is a file

④ Make a script

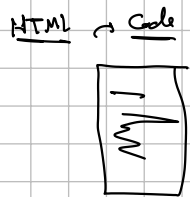
⑤ make each program as Filter

⑥ Avoid Captive GUI

⑦ Don't reinvent the wheel

⑧ Make Proto Type ASAP

Every thing is File



What is the File Types:-

① Regular File (File.txt) we can generate $\begin{matrix} \swarrow \text{UI Tool} \\ \text{Command (script)} \\ \searrow \text{Code} \end{matrix}$ Touch

② Directories (1st set of files)

③ Device & Peripheral (/dev) → Read (write)
→ stdin
→ stdout
→ stderr

④ Pipe

⑤ Socket → InterProcess (devices) Communication

⑦ Links (symbolic) (shortcut) $\begin{matrix} \text{File} \leftarrow \text{meta} \\ \text{body} \rightarrow \end{matrix}$ Binary address (Pointer) File Pointer

abstract
C →
Posix
→ Autosar
→ CMSIS (ARM)
→ Adaptive Autosar

File Name:-

→ Case sensitive (test) (TEST)

→ name length (unlimited) 255/32

→ extension → File.txt | File.mps

File.txt.old →
File.txt →
makefile

→ ①

→ index.html

extension
Name $\begin{matrix} \text{extension} \\ \text{indicator} \end{matrix}$
- File Type
- know which APP to use

→ Separator → Path

/ test / APP

In → escape se
/ Path

* Path *

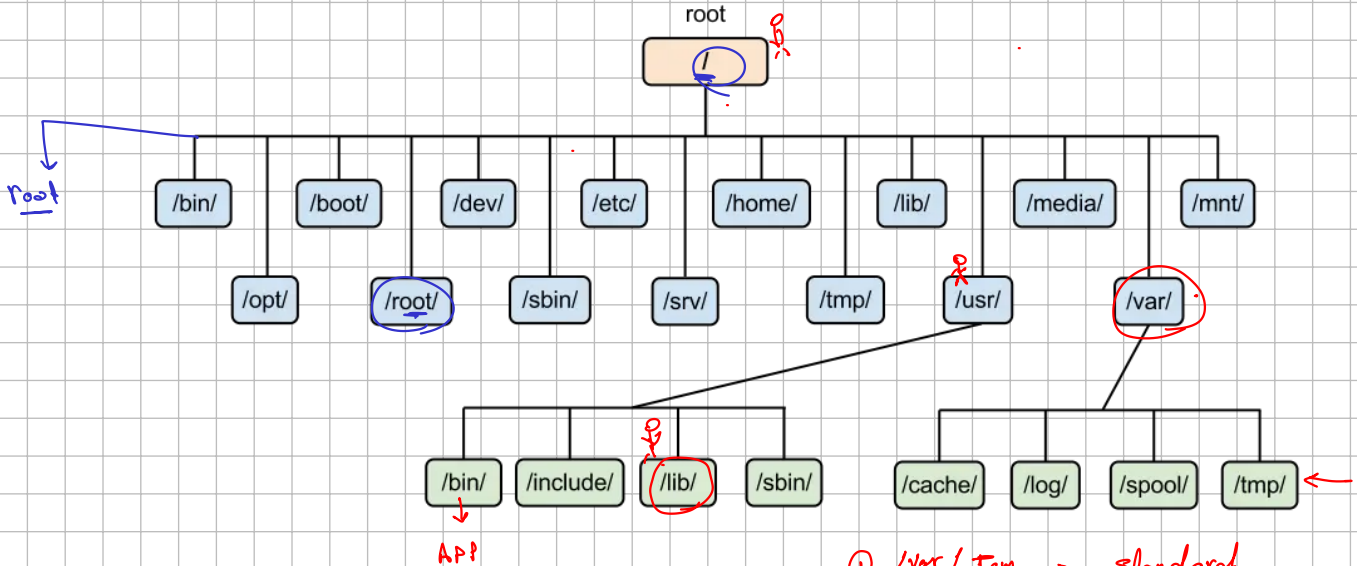
touch

Path → is sequence of nested directory with File or directory @ the end,

Separated by /

Path

Absolute Path:- Path From root to the end of tree



① /var/tmp → standard

... /var/tmp

./log
Current

Relative:- to current

- current directory
- Parent directory

#include "relative"

Toolchain → about

① root (root)

- Shell:- tools to execute command

- why called shell:- because they hide details happens in OS

- Command Code (APP) [Input in text terminal] →

- Results of command displayed as text,

- Shell can execute scripts

well

sh:- traditional, basic shell found on unix by Steve Bourne.

bash:- born again shell

csh, tsh, ksh, zsh, fish, etc.

~ → telah (home directory)
/home/user/eslam

Command Format

Name [argument] [option]
↳ Selection

Special directory
. / / .. / ~ /

Note

~ / → home directory

Shell substitutes with → /home/user/eslam

Don't use it in Programs

Operation can be done on file

- ✓ ① creat ② delete_{rm} ③ copy_{cp} ④ move_{mv} ⑤ display ⑦ edit ⑧ change file permission

⑨ display Path

* Directory *

- ① creat ② delete_{rm} ③ copy, move, ④ display / change ⑤ change permission

① creat file

Touch

Touch { 0 = 100 } 100 0 → 99
{ 0, 100 } 2

②

cat /home/oracle/scripts/start_all.sh <<EOF

#!/bin/bash

./home/oracle/scripts/setEnv.sh

export ORAENV_ASK=NO

. oraenv

export ORAENV_ASK=YES

dbstart \ \$ORACLE_HOME

EOF

Note:-

<https://ubuntu.com/tutorials/command-line-for-beginners#1-overview>

Command Documentation

① --help →

② man →

③ w.kel

① Touch → creat

② redirection → outPut → store



Redirection:-

Redirection permits a user to alter from the defaults for stdin, stdout, or stderr to a device or file using their file descriptors

File structure:-

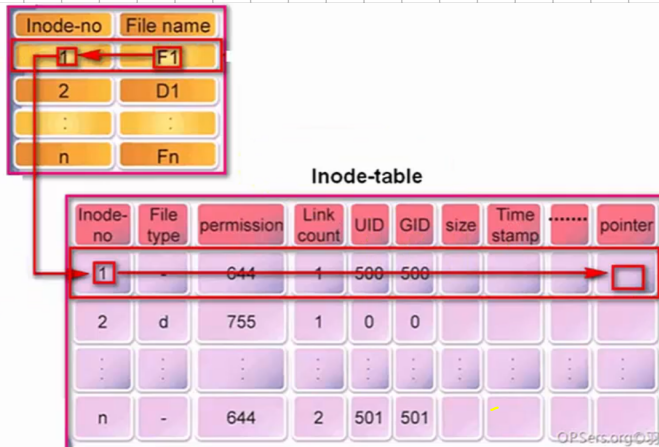
1. inode → struct (Metadata)

2. Payload (data)

- Type, Permission, owner ID, Group ID, size, last access time

- reference (Pointer) → (Block) beginning address

→ ls -li
→ stat



```

ls -la /home/eslam/
total 129344
-rw-rw-r-- 1 eslam eslam 283002 Jan 17 20:23 2026-01-17-Note-20-23.xopp
-rw-rw-r-- 1 eslam eslam 923230 Jan 17 22:58 2026-01-17-Note-20-23_annotated.pdf
drwxr-xr-x 2 eslam eslam 4096 Jan 23 18:45 Desktop
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:18 Documents
drwxr-xr-x 4 eslam eslam 4096 Jan 23 20:40 Downloads
drwxrwxrwx 21 eslam eslam 4096 Jan 23 18:59 [redacted]
drwxrwxr-x 2 eslam eslam 4096 Jan 23 20:51 LinuxFundamentals
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:18 Music
drwxr-xr-x 3 eslam eslam 4096 Jan 17 20:35 Pictures
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:18 Public
drwxr-xr-x 2 eslam eslam 4096 Jan 18 01:25 Templates
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:18 Videos
-rw-rw-r-- 1 eslam eslam 0 Jan 23 19:36 app
drwxrwxr-x 5 eslam eslam 4096 Jan 18 01:28 claude-desktop-debian
-rw-r--r-- 1 eslam eslam 135266304 Jan 17 19:38 core.28063
-rw-r--r-- 1 eslam eslam 98607104 Jan 17 19:39 core.28507
-rw-rw-r-- 1 eslam eslam 0 Jan 23 20:15 f1
-rw-rw-r-- 1 eslam eslam 0 Jan 23 20:15 f2
-rw-rw-r-- 1 eslam eslam 0 Jan 23 20:15 f3
-rw-rw-r-- 1 eslam eslam 119196224 Jan 17 11:43 google-chrome-stable_current_and64.deb
-rw-rw-r-- 1 eslam eslam 0 Jan 23 19:37 napp
drwx----- 13 eslam eslam 4096 Jan 18 01:47 snap
drwxrwxr-x 2 eslam eslam 4096 Jan 23 19:22 test
drwxrwxr-x 2 eslam eslam 36864 Jan 23 20:19 testdir
-rw-rw-r-- 1 eslam eslam 24 Jan 23 20:25 text.f
-rw-rw-r-- 1 eslam eslam 24 Jan 23 20:25 text.y
    
```

- Read write excute



111
rwx rwx rwx
111 111 100
7 3 4

7 5 5
111 101 101
6 4 4
110 100 100

Redirection

stdin Redirection <

nc < text .

Script

Here docs " " <<

• open question

<<<

> stdout redirection

Whaami > F.txt

>> append

Screen

→ echo "File.txt" > File.txt

By default stdout

echo "File.txt"

1 > File.txt

overwrite

2 > File.txt

2 >>

Append

ls /xyz

2 > /dev/null

Terminal (Console) →

Screen

Keyboard

(file) Input
stdin

Shell

output

→ stdout = 1

→ stderr = 2

→ Combine

echo "string" 1> & 2>

echo " " 1> & 2>

ls /etc /xyz 2> err 1> & 2

Not for

err

Pipe |

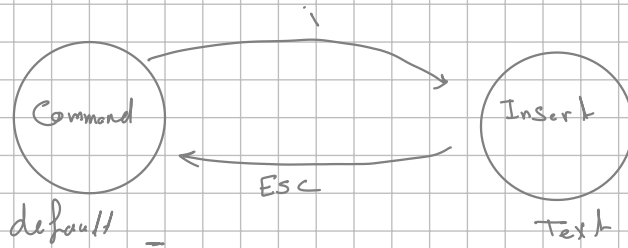
Connect stdout of one Process to stdin another Process

Note:-
Short cut

term zero & ?
1 &

|

③ Text editor
① nano



① Arrow **h (←)** **k (↑)** **L (→)** **J (↓)**

i → insert
a → append

G → go last line

5G → go line no. 5
Ctrl + G →

② Ctrl + f → one page forward
Ctrl + b → one page backward

selection

shift + Arrow → letter

shift + ~~ctrl~~ + Arrow → word

shift ↑
↓ → statement